revision for linear independence

for every $v \in \text{span}\{x_1, \ldots, x_n\}$ the equation $v = a_1x_1, \ldots, a_nx_n$ has exactly one solution

---

Ex: what is the dim of $\mathbb{C}$ as a vector space over $\mathbb{R}$?

number of elements in a basis

a basis for $\mathbb{C}$ as a vector space over $\mathbb{R}$ is a collection of vectors $\{c_1, \ldots, c_n\} \subset \mathbb{C}$ such that

1) $\text{span}\{c_1, \ldots, c_n\} = \mathbb{C}$

$\{a_1c_1, \ldots, a_nc_n \mid a_i \in \mathbb{R}\}$

2) $\{c_1, \ldots, c_n\}$ is linearly independent

Conj: $\{1, i\}$ is a basis?

does $\{1, i\}$ span $\mathbb{C}$? can I write any complex # $c \in \mathbb{C}$ as a linear combination of $\{1, i\}$? are there $a, b \in \mathbb{R}$ s.t. $c = a1 + bi$? yes

well $\mathbb{C} := \{a + bi \mid a, b \in \mathbb{R}\}$

is $\{1, i\}$ linearly independent? yes cuz our scalars are from $\mathbb{R}$

so $\{1, i\}$ is a basis $\underbrace{|\{1, i\}| = 2}_{\text{dim } 2}$

dimension of a vector space depends on the field of scalars

Ex: What is dim P_2? (Scalars from $\mathbb{R}$)

basis: $\{(x^2, x, 1)\}$ dim 3 :)

$P_2 := \{ax^2+bx+c(1) \mid a,b,c \in \mathbb{R}\} = \text{span}\{x^2, x, 1\}$

What is dim P_n? $n+1$

$\{x^n, \dots x^1, 1\}$ span ✓ linearly independent ✓

Ex: What is dim $\{\text{differentiable functions } \mathbb{R} \to \mathbb{R}\}$?

$P_n \subset$ {differentiable functions} $\ni x \mapsto \sin x,\ x \mapsto e^x$

Since $P_{2025} \subset \{\text{diff func } \mathbb{R} \to \mathbb{R}\}$ $\dim P_{2025} \leq \dim\{\text{diff func } \mathbb{R} \to \mathbb{R}\}$

$\forall n$ $P_n \subset \{\text{diff func } \mathbb{R} \to \mathbb{R}\}$, $n+1 = \dim P_n \leq \dim\{\text{diff func } \mathbb{R} \to \mathbb{R}\}$

We're implicitly using

Thrm: if $W \subset V$ and W is a subspace of V then

$\dim W \leq \dim V$

(if dim W is finite)

$\Rightarrow$ $\dim\{\text{diff func } \mathbb{R} \to \mathbb{R}\}$ is not finite

if V and W have finite bases, then proof by contradiction

if dim V is not finite and dim W is finite ✓

Not all vector spaces are finite dimensional